HL7 Documentation

I3SFU_HL7_Documentation_06

Revision number	Description	Author	Effective Date
1	Initial version	Csikós Jenő	2014/10/10
2	HL7 PID added, Flags added, non-printable characters amended, add stream (txt file)	Magyar Csaba	2015/01/15
3	Reagent Lot added, Display range added	Csikós Jenő	2015/02/11
4	HL7 message version	Tremmel Attila	2015/03/04
5	HL7 new message version	Tremmel Attila	2015/06/08
6	Bidirectional HL7, new Hl7 message format, flags update	Tremmel Attila	2019/05/22

Approver Name	Position	Signature	Date
Csikós Jenő	Software Developer		2019/05/22
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HL7 Minimum Layer Protocol (MLP)

Icon software has the **HL7 v2.5** protocol implemented for transmission of results.
The HL7 message goes through a TCP/IP connection, according to the MLP (Minimal Layer Protocol). As the HL7 message does not have a predefined length, therefore there is a frame, one byte at the beginning and two bytes at the end; this way the receiver will be able to detect the different messages. These headers and trailers are usually non-printable characters that would not typically be in the content of (printed) HL7 messages.

The header is a vertical tab character <VT> (hex 0x0b). The trailer is a file separator character <FS> (hex 0x1c) immediately followed by a carriage return <CR> (hex 0x0d).
It will look like this:

<VT> **(hex 0x0b)**	**HL7 Message goes here**	**<FS>** **(hex 0x1c)**	**<CR>** **(hex 0x0d)**

Components of an HL7 Message

Each HL7 standard message is composed of groups and segments where:

- Groups contain segments or groups
- Segments contain fields
- Fields contain subfields

ORU R01 Message

The message type sent by the Analyzer is ORU R01.
The first segment (MSH) looks like as follows:

```
MSH|^~\&|AnalyzerName|ND30H14003|LISApplication|LIS|20090501002750||ORU^R01|63376
7344705480000|P|2.5||||||UNICODE UTF-8
```

Fields of Message Header (MSH) Segment:

- Name (MSH)
- Field Separator (|)
- Encoding Characters (^~\&)
- Sending Application (AnalyzerName)
- Sending Facility (ND30H14003)
- ...

- Message Type (ORU^R01)
  - Message Code (ORU)
  - Trigger Event (R01)
- etc.

In a segment the fields are separated by the Field Separator (except Field Separator component), subfields are separated by the first encoding character internally.

Each line (field) is terminated with <CR><LF> (hex 0x0d 0x0a).

Transmission of additional data (flags)

Icon is able to include technical flags with the reports. These flags are included as notes (NTE) and are named as PLTFlags, WBCFlags and RBCFlags.

Each flag character is followed by a number indicating the “weight” or the “severity” of the error. If a flag has a “weight” equal to or higher than 6, then corresponding parameters are dashed out.

```
…
NTE|Comment4||x3|4^RBC flags
NTE|Comment5||p5|5^PLT flags
NTE|Comment6||X4N6|6^WBC flags
…
```

* This NTE format is valid only for Format A, Format B and Format C

The interpretation of (highlighted) characters is listed below:

WBC Flags		RBC Flags		Technical Flags	
A	Unstable Voltage	a	Unstable Voltage	B	Low sample volume
C	3part fallback	g	Distorsion	O	Open mode blood sensor disabled
D	WBC channel dirty	H	Unstable HGB		
G	Distorsion	h	Unstable HGB baseline		
L	Range Exceeded	l	Range Exceeded		
N	Noise	n	Noise		
V	Improper voltage	v	Improper voltage		
W	Unintelligible Histogram	x	Overload		
X	Overload				
Y	Inadequate Lysis				
Z	WBC fallback				

Note: all flag texts depend on the local language. Icon will only transmit flag letters. For interpretation and resolution of flags, please consult the Operator’s Manual of the analyzer.

Brief segment table

MSH	
1	FieldSeparator
2	EncodingCharacters
3	SendingApplication
4	SendingFacility
5	ReceivingApplication
6	ReceivingFacility
7	DateTimeOfMessage
8	Security
9	MessageType
10	MessageControlID
11	ProcessingID
12	VersionID
13	SequenceNumber
14	ContinuationPointer
15	AcceptAcknowledgmentType
16	ApplicationAcknowledgmentType
17	CountryCode
18	CharacterSet
19	PrincipalLanguageOfMessage
20	AlternateCharacterSetHandlingScheme
21	MessageProfileIdentifier

SFT	
1	SoftwareVendorOrganization
2	SoftwareCertifiedVersionOrReleaseNumber
3	SoftwareProductName
4	SoftwareBinaryID
5	SoftwareProductInformation
6	SoftwareInstallDate

OBX	
1	SetIDOBX
2	ValueType
3	ObservationIdentifier
4	ObservationSubID
5	ObservationValue
6	Units
7	ReferenceRange
8	AbnormalFlags
9	Probability
10	NatureOfAbnormalTest
11	ObservationResultStatus
12	EffectiveDateOfReferenceRange
13	UserDefinedAccessChecks
14	DateTimeOfTheObservation
15	ProducersID
16	ResponsibleObserver
17	ObservationMethod
18	EquipmentInstanceIdentifier
19	DateTimeOfTheAnalysis

OBR	
1	SetIDOBR
2	PlacerOrderNumber
3	FillerOrderNumber
4	UniversalServiceIdentifier
5	PriorityOBR
6	RequestedDateTime
7	ObservationDateTime
8	ObservationEndDateTime
9	CollectionVolume
10	CollectorIdentifier
11	SpecimenActionCode
12	DangerCode
13	RelevantClinicalInformation
14	SpecimenReceivedDateTime
15	SpecimenSource
16	OrderingProvider
17	OrderCallbackPhoneNumber
18	PlacerField1
19	PlacerField2
20	FillerField1
21	FillerField2
22	ResultsRptStatusChngDateTime
23	ChargeToPractice
24	DiagnosticServerSectID
25	ResultStatus
26	ParentResult
27	QuantityTiming
28	ResultCopiesTo
29	ParentNumber
30	TransportationMode
31	ReasonForStudy
32	PrincipalResultInterpreter
33	AssistantResultInterpreter
34	Technician
35	Transscriptionist
36	ScheduledDateTime
37	NumberOfSampleContainers
38	TransportLogisticsOfCollectedSample
39	CollerctorsComment
40	TransportArrangementResponsibility
41	TransportArranged
42	EscortRequired
43	PlannedPatientTransportComment
44	ProcedureCode
45	ProcedureCodeModifier
46	PlacerSupplementalServiceInformation
47	FillerSupplementalServiceInformation
48	MedicallyNecessaryDuplicateProcedureReason
49	ResultHandling

NTE	
1	SetID
2	SourceOfComment
3	Comment
4	CommentType

Message

The message consists of MSH, SFT, OBR, NTE and OBX segments.

- Message Header (MSH) contains message relevant information
- Software segment (SFT) contains sender software information
- Observation Request (OBR) contains vial information (sample id)
- Notes and Comments (NTE) contains notes and comments to the segment followed by the NTE
- Observation Result (OBX) contains individual result of a given parameter

The below message has all non-printable characters removed. All lines have <CR><LF> removed from its end.


```
MSH|^~\&|Icon|NI1400010 |LIS
Application|LIS|20150209030529||ORU^R01|635590479295400000|P|2.5||||||UNICODE UTF-8
SFT|Custom Distributor|0.2.413.0|Icon|0.2.413.0|Product Version: 0.9 Software complete version: 0.2.413.0(FE - 00 -
17)|20150205070004
OBR||||^^^400^23456||||20150116005609|25||||3 Part Differential Hematology
NTE|Comment1|||1^Name
NTE|Comment2|||2^Age
NTE|Comment3|||3^Comment
NTE|Comment4|||4^RBC flags
NTE|Comment5|||5^PLT flags
NTE|Comment6|||6^WBC flags
NTE|Comment7||B3|7^Technical flags
NTE|Comment8||91715010614_00010|8^Reagent Diluent
NTE|Comment9||91715010614_00010|9^Reagent System Solution
NTE|Comment10||91715010614_00010|10^Reagent Lyse
OBX|||0^RBC||0.00|^10⁶/µL|3.88-5.78||||F
OBX|||1^HGB||0|^g/L|120-172||||F
OBX|||2^MCV||<30|^fL|78-96||||F
OBX|||3^HCT||0|^%|35-51||||F
OBX|||4^MCH||0.00|^pg|26.40-33.20||||F
OBX|||5^MCHC||0|^g/L|318-367||||F
OBX|||6^RDWsd||<1.0|^fL|<1.0-<1.0||||F
OBX|||7^RDWcv||<1.0|^%|11.3-14.7||||F
OBX|||8^PLT||0|^10³/µL|172-440||||F
OBX|||9^MPV||0.0|^fL|6.1-9.1||||F
OBX|||10^PCT||0.00|^%|0.00-0.00||||F
OBX|||11^PDWsd||<1.0|^fL|<1.0-<1.0||||F
OBX|||12^PDWcv||<1.0|^%|<1.0-<1.0||||F
OBX|||13^WBC||0.01|^10³/µL|3.70-11.70||||F
OBX|||14^LYM||-|^10³/µL|1.10-3.60||||F
OBX|||15^LYMP||-|^%|14.1-52.8||||F
OBX|||16^MID||-|^10³/µL|0.20-1.20||||F
OBX|||17^MIDP||-|^%|3.2-17.7||||F
OBX|||18^GRA||-|^10³/µL|1.90-7.90||||F
OBX|||19^GRAP||-|^%|39.6-78.4||||F
OBX|||20^PLCR||0|^%|0-0||||F
OBX|||21^PLCC||0|^10³/µL|0-0||||F
OBX|||22^RBCHistogram||AAABBAYIBwYFAwEAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
```

```
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAA==||||||F
NTE|RD||27|1^RBC Discriminator (fL)
OBX|||23^PLTHistogram||AAAAAAAAAAAAAAAAAAABAgMEBQYHCAgHBgUEAwIBAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAA==||||||F
OBX|||24^WBCHistogram||AAAAAAAAAAAAAAAAAAAAAAAAAAEAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAA==||||||F
NTE|WD1||0|1^WBC Discriminator #1 (fL)
NTE|WD2||0|2^WBC Discriminator #2 (fL)
```


Note 1: RBCHistogram, PLTHistogram WBCHistogram values are Base64 string encoded 256 length byte arrays.

Note 2: The PID field includes (repeats) the sample identification string. This feature must be enabled in Icon operating software (refer to Operator's Manual).

```
…
PID||| ||23456
…
```

Note 3: Observation Values of OBX segments can contain '-', '<' or '>' characters in case of non-evaluated result or if the newly introduced display limitations were triggered. For example if MCV value exceeds the lower display limit (30 fL) than the Observation value is going to be: <30

```
…
OBX|||2^MCV||<30|^fL|78-96||||F
…
OBX|||14^LYM||-|^10³/µL|1.10-3.60||||F
…
```

Note 4: Comment value of NTE segment – where Text value of Comment type is Reagent Diluent, Reagent System Solution or Reagent Lyse - contains the LOT identifier of the active reagent during Measurement.

```
…
NTE|Comment8||91715010614_00010|8^Reagent Diluent
NTE|Comment9||91715010614_00010|9^Reagent System Solution
NTE|Comment10||91715010614_00010|10^Reagent Lyse
…
```

Note 5: Comment value of NTE segment – where Text value of Comment type is QC LOT – contains the QC Lot of the given QC measurement.

```
…
NTE|Comment11||1N0301N|11^QC LOT
…
```

Acknowledgment

Receiver application should reply to the message before connection is closed. In other words sender application tries to wait until the acknowledgment arrives after that sender initiates the closing of connection. Awaiting time is 9 seconds. Typical acknowledgment is the following:

```
MSH|^~\&|||AnalyzerName|ND30H14003|20140430150042||ACK|633767344705480000|P|2.5
MSA|AA|635590479295400000
```

Content of Message Header (MSH):

- AnalyzerName is now the Receiving Application
- ND30H14003 is now the Receiving Facility
- ACK: the type of the message
- 633767344705480000 Message Control ID

Content of Message Acknowledgment (MSA):

- AA Acknowlegment Code
    - AA Application Accept
    - AE Application Error
    - AR Application Reject
- 635590479295400000 Message Control ID which this Acknowledgment belongs to.

Note: A typical transmission stream should accompany this document (UTF-8 text file) to support the implementation of a receiver tool.

Versions

To achieve backward compatibility in software version tracking is introduced. Software segment (**SFT**) contains a field *SoftwareProductInformation*, which is responsible to report message format. The following formats are available:

- Format A
  - mentioned field starts with “Product Version:” string to ensure backward compatibility
- Format B
  - mentioned field starts with “Message Version: Version_0_2_437” string
- Format C
  - mentioned field starts with “Message Version: Version_0_2_480” string
- Format D
  - mentioned field starts with “Message Version: Version_0_4_929” string

Main attributes of the Format A message are:

- *ObservationValue* field of OBX segments can contain ‘-‘ character
- *SoftwareProductInformation* field of Software segment starts with “Product Version:” string
- QC LOT value is sent in *SourceOfComment* NTE field with Comment8 value

Main attributes of the Format B message are:

- *ObservationValue* field of OBX segments can contain ‘-‘, ‘<’, ‘>’ character
- *SoftwareProductInformation* field of Software segment starts with “Message Version: Version_0_2_437” string
- NTE segment contains the LOT identifier of the active reagents during measurement

Main attributes of the Format C message are:

- Based on Format B message
- *SoftwareProductInformation* field of Software segment starts with “Message Version: Version_0_2_480” string
- NTE segment contains the WD0 identifier as WBC Discriminator #0 before WD1 identifier

Main attributes of the Format D message are:

- Based on Format C message
- *SoftwareProductInformation* field of Software segment starts with “Message Version: Version_0_4_929” string
- NTE segment data format is different
- NTE segment contains the profile

* This NTE format is valid only for Format D

```
…
NTE|Comment1|||RE^Name
NTE|Comment2|||RE^Age
NTE|Comment3|||RE^Comment
NTE|RBC flags||a3
```

```
NTE|PLT flags
NTE|WBC flags||A3
NTE|Technical flags||O3
NTE|Reagent Diluent||91711650417_00064
NTE|Reagent System Solution||91711650417_00064
NTE|Reagent Lyse||91711650417_00064
NTE|Profile||Human
…
NTE|RD||36|RE^RBC Discriminator (fL)
…
NTE|WD0||32|RE^WBC Discriminator #0 (fL)
NTE|WD1||65|RE^WBC Discriminator #1 (fL)
NTE|WD2||101|RE^WBC Discriminator #2 (fL)
…
```

NTE segment CommentType field ***optionally*** contains additional information in Format D message. E.g. NTE segment of Comment1 as SetID field contains the user defined value of Comment1 data.

Bidirectional HL7

Normal order

```
MSH|^~\&|LIS||LIS||20190514101754||ORM^O01|MSG-20190514-101754|P|2.5
ORC|NW
OBR||||SAMPLE_HUMAN||20190514152124
NTE|Comment1||Name
NTE|Comment2||Age
NTE|Comment3||Comment
NTE|Profile||Human
```

Cancel order

```
MSH|^~\&|LIS||LIS||20190514101755||ORM^O01|MSG-20190514-101755|P|2.5
ORC|CA
OBR||||SAMPLE_HUMAN||20190514152125
NTE|Comment1||Name
NTE|Comment2||Age
NTE|Comment3||Comment
NTE|Profile||Human
```

Using ORM^O01 LIS application could send a Normal order or Cancel order to analyzer. To accept a new normal order the analyzer should not contain an existing normal order with the same properties:

- Sample ID
- Requested/Ordered Date and Time (yyyyMMddHHmmss)
- Comment1

- Comment2
- Comment3
- Profile (Human, Male, Female, LT4, LT5, LT6, LT7, LT8)

To delete a normal order the analyzer should contain an existing normal order with the same properties:

- Sample ID
- Requested/Ordered Date and Time (yyyyMMddHHmmss)
- Comment1
- Comment2
- Comment3
- Profile (Human, Male, Female, LT4, LT5, LT6, LT7, LT8)

Accepted order acknowledgement

```
MSH|^~\&|AnalyzerName||MachineID||20190514101756||ACK|MSG-20190514-101756|P|2.5
MSA|AA|MSG-20190514-101754
```

Rejected order acknowledgement

```
MSH|^~\&|AnalyzerName||MachineID||20190514101756||ACK|MSG-20190514-101756|P|2.5
MSA|AR|MSG-20190514-101754
ERR|||207^Order already exists|E
```

Analyzer sends acknowledgment (ACK) message to any valid HL7 v2.5 message. This message contains MSA segments where *AcknowledgementCode* field contains ‘AA’ or ‘AR’ string. ‘AA’ means order is accepted; ‘AR’ means order is rejected. In some cases analyzer sends ERR segment in ACK message to explain ‘AR‘ *AcknowledgementCode.*